

ENEE452 HW07 - Problem 1

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Critical Region 1 (Lines 71-75)

If link A is freed and msgPost() interrupts before the next line in the msgSend(), msgPost() will find link A free to write new message data into and append A to the tail. Then when msgSend() resumes, the links are misordered in the list. This can create a duplicate output of A.

Critical Region 2 (Lines 112-114)

If the listheadp write is performed and then interrupted by msgSend(), it would find bytecount still 0 even though it shouldn't be, as the byteptr and bytecount writes have not been performed yet. This means the first link can be skipped.

Critical Region 3 (Lines 117-118)

The old tail's next pointer must be chained before listtailp is updated. If not, msgSend() will determine that the list is exhausted and frees the old tail, and the link is lost.

Critical Region 4 (Line 121)

If msgPost() could be called in two different instances simultaneously, the window between the listtailp = pLink write and next = pLink write allows the second caller to see next == NULL and begin populating the same link for a different message. This would corrupt the current list.